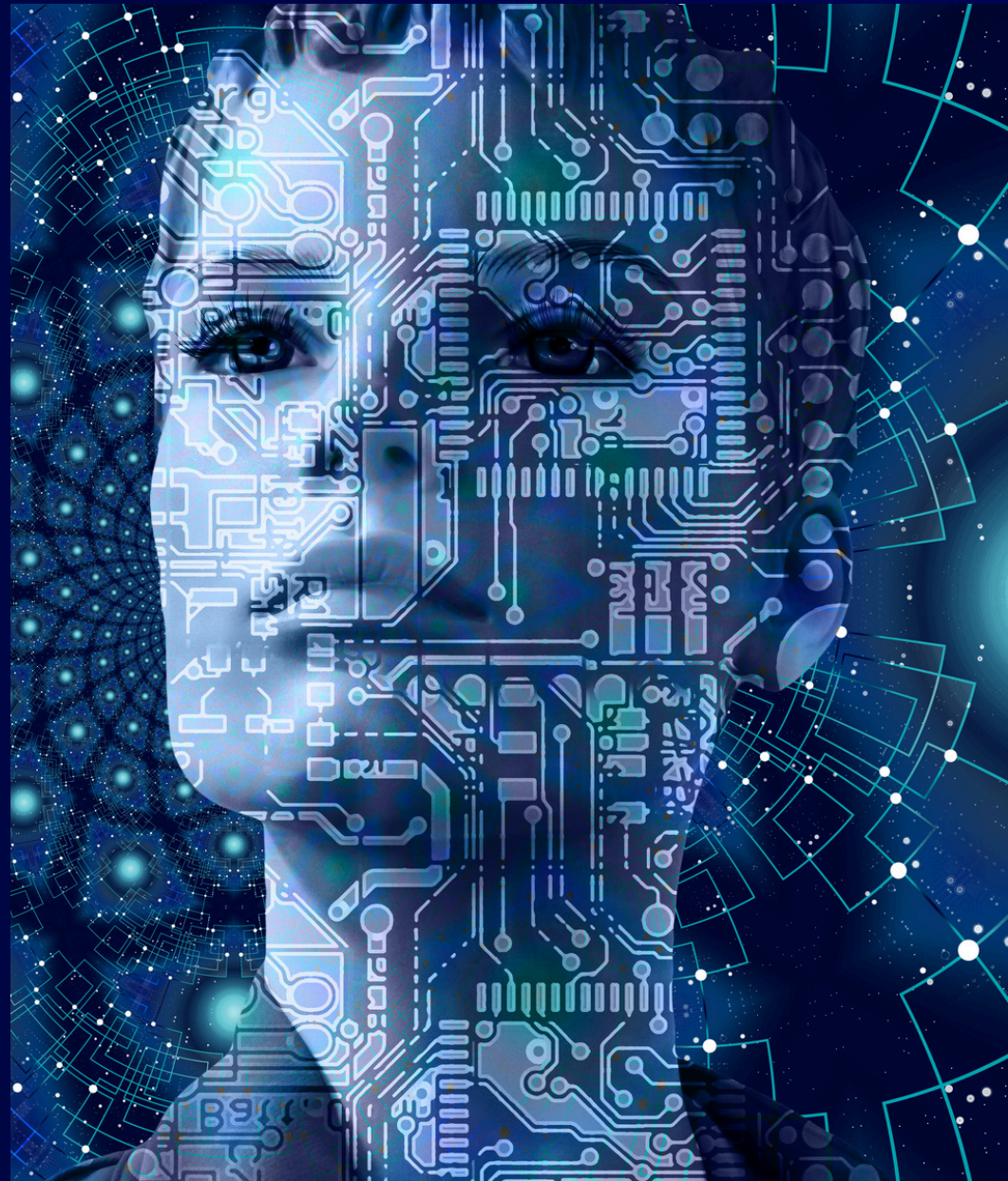


DATA ANALYSIS PROJECTS

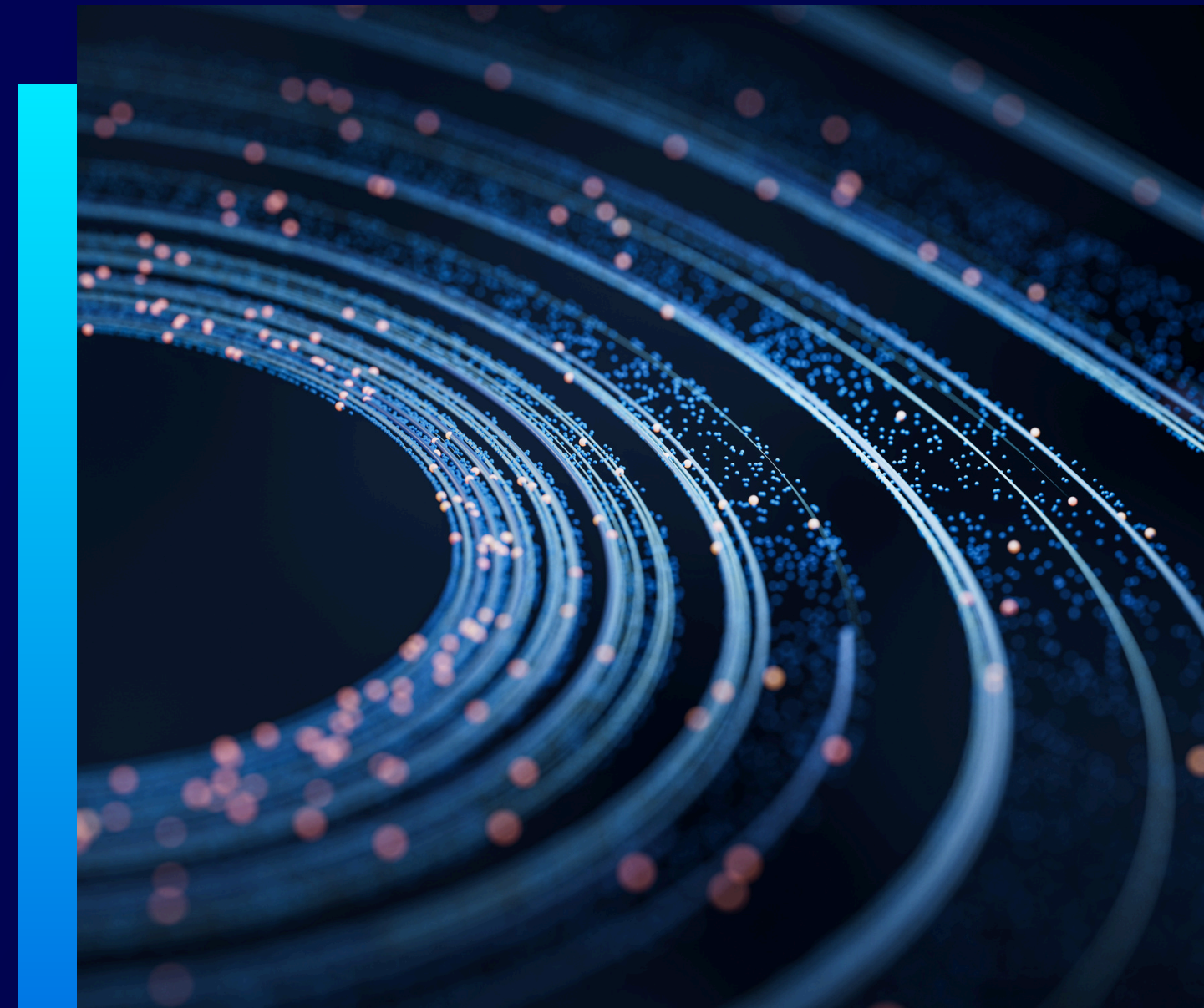


THE IMPORTANCE OF DATA ANALYSIS

In today's digital age, data is a critical asset. Effective data analysis enables organizations to identify trends, improve operations, and discover new opportunities. It bridges the gap between data collection and strategic implementation.

PROJECT

**E-COMMERCE FURNITURE SALES DATASET 2024
ELECTRIC VECHILE SALES BY STATE IN INDIA
GOOGLE PLAY STORE APPS
OLYMPIC SUMMER MEDAL FROM 1976-2008**



E COMMERCE FURNITURE SALES DATA SET SUMMARY

PROJECT OVERVIEW

TITLE: E-COMMERCE FURNITURE SALES

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

**DATASET: CONTAINS 2,000 ALIEXPRESS FURNITURE PRODUCT ENTRIES WITH
SALES AND PRODUCT DETAILS.**

PROJECT OBJECTIVE

Main Goal: Predict the number of furniture items sold based on features like product title, original price, current price, and tags.

DATASET DETAILS

FIELDS INCLUDE:

- PRODUCTTITLE
- ORIGINALPRICE
- PRICE
- SOLD
- TAGTEXT (E.G., "FREE SHIPPING").

USEFUL FOR: EXPLORATORY DATA ANALYSIS, SALES
PREDICTION, MARKET TRENDS, IMPACT OF
DISCOUNTS, PRODUCT POPULARITY ANALYSIS.
DATA COLLECTED ETHICALLY FROM ALIEXPRESS

```
*** Data Loaded Successfully!
```

```
Shape of Dataset: (2000, 5)
```

```
Preview of Data:
```

	productTitle	originalPrice	price	\
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	\$30	\$46.79	
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	\$116	\$169.72	
2	Desser For Bedroom With 7 Fabric Drawers Organ...	\$78.4	\$39.46	
3	Modern Accent Boucle Chair,Upholstered Tufted ...	\$89	\$111.99	
4	Small Unit Simple Computer Desk Household Wood...	\$48.82	\$21.37	

	sold	tagText
0	600	Free shipping
1	9	Free shipping
2	7	Free shipping
3	2	Free shipping
4	1	Free shipping

METHODOLOGY

1. **Data Collection: Dataset loaded via pandas as a CSV file.**
2. **Data Preprocessing:**
 - **Handling missing values (dropping incomplete rows)**
 - **Encoding categorical variables (tagText and productTitle via LabelEncoder and TF-IDF)**
 - **Feature engineering (e.g., calculating percentage discount).**

...

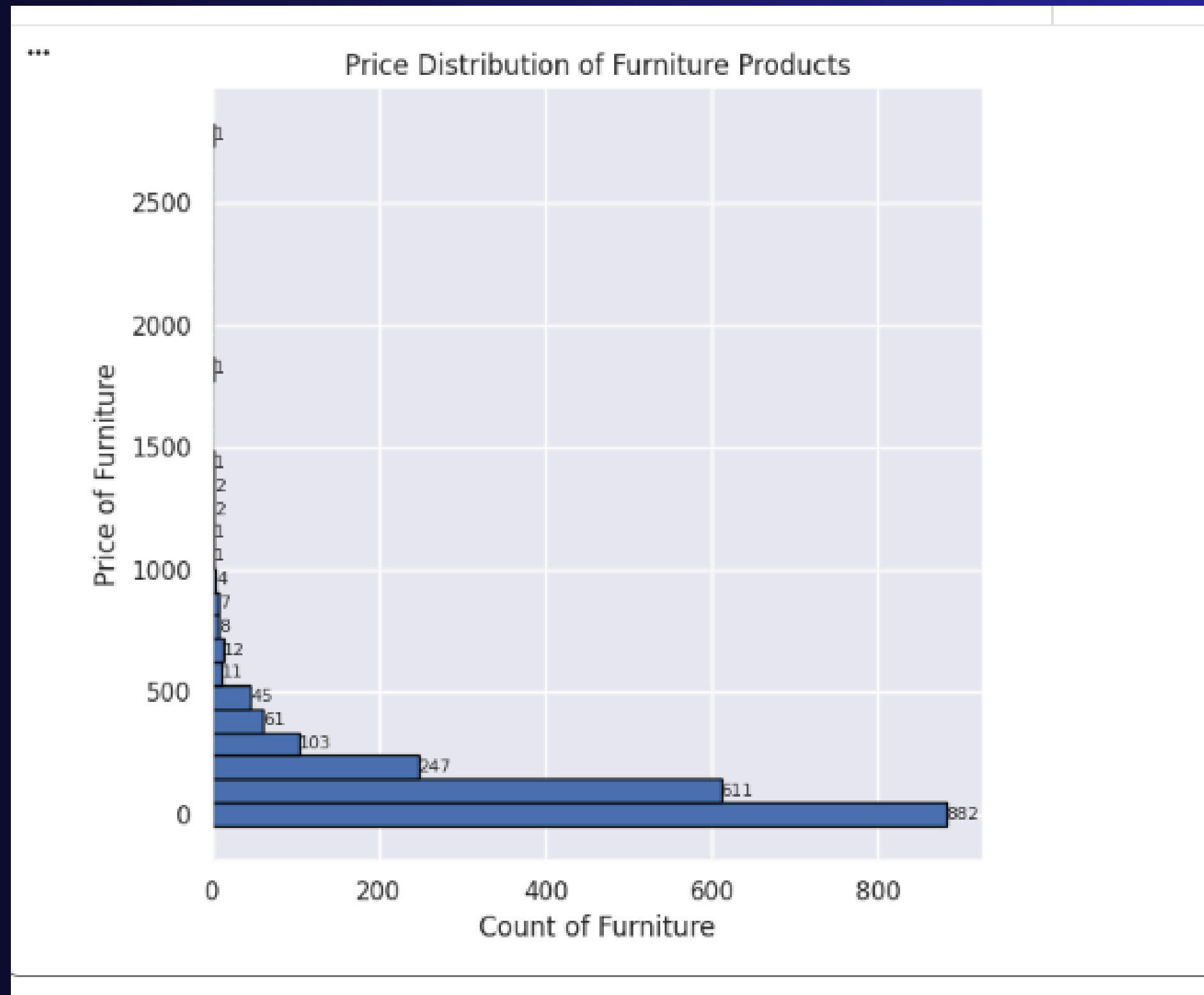
✅ Cleaned Data Sample:

	productTitle	originalPrice	price	\
0	Dresser For Bedroom With 9 Fabric Drawers Ward...	30.00	46.79	
1	Outdoor Conversation Set 4 Pieces Patio Furnit...	116.00	169.72	
2	Desser For Bedroom With 7 Fabric Drawers Organ...	78.40	39.46	
3	Modern Accent Boucle Chair,Upholstered Tufted ...	89.00	111.99	
4	Small Unit Simple Computer Desk Household Wood...	48.82	21.37	

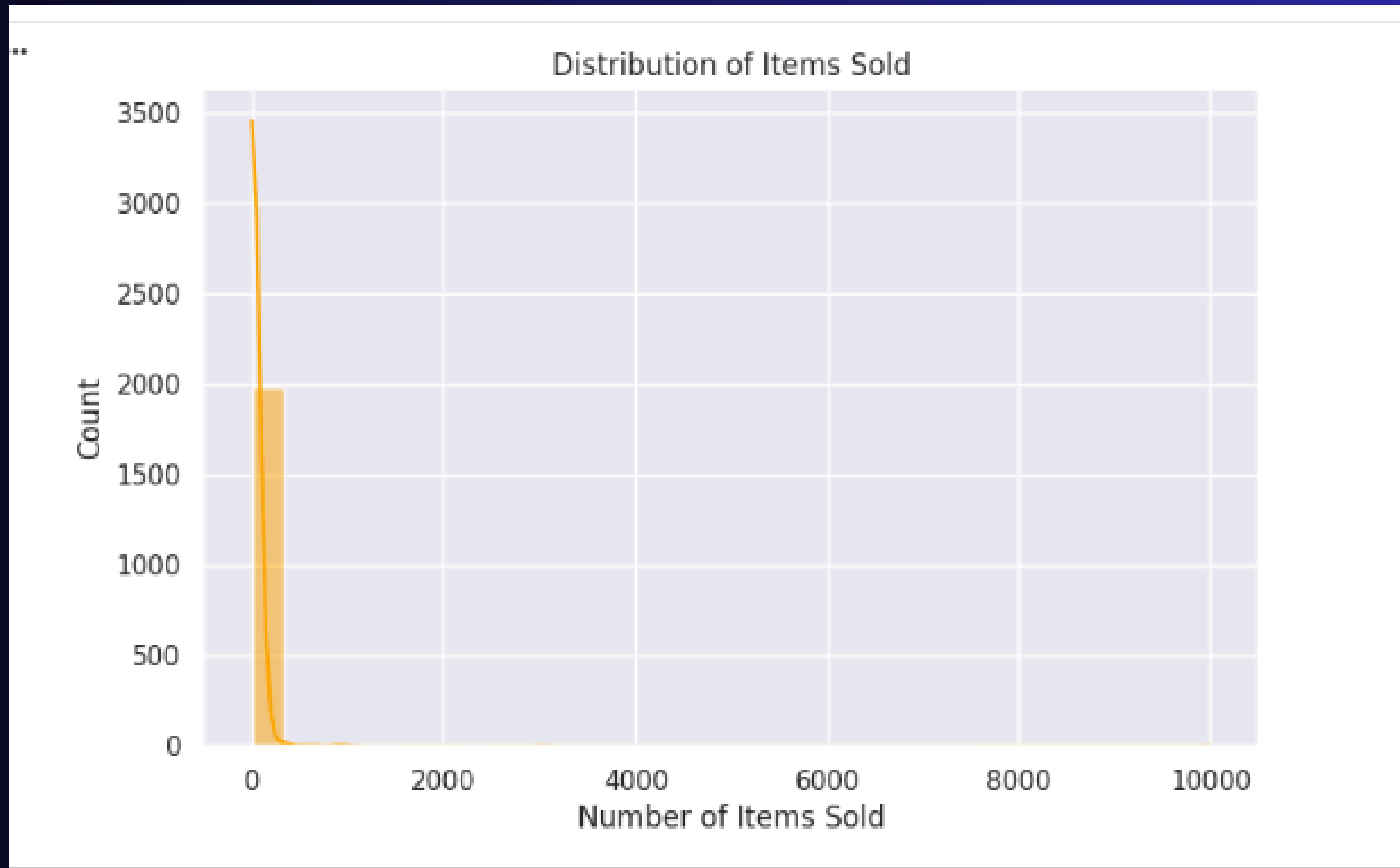
	sold	tagText
0	600	Free shipping
1	9	Free shipping
2	7	Free shipping
3	2	Free shipping
4	1	Free shipping

EXPLORATORY DATA ANALYSIS

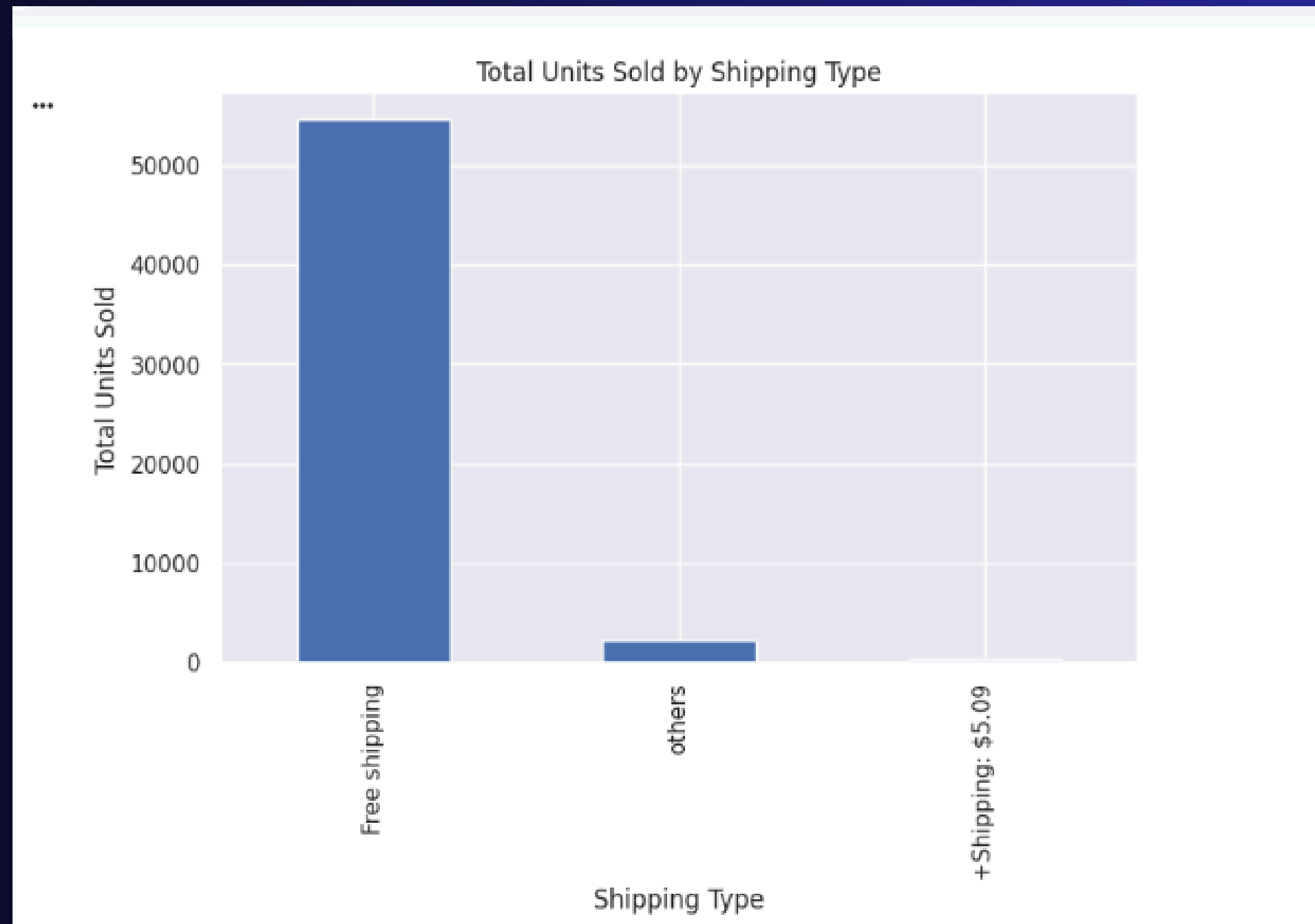
PRICE DISTRIBUTION OF FURNITURE



DISTRIBUTION OF ITEMS SOLD



PRICE VS SOLD FURNITURE



SHIPPING OPTION DISTRIBUTION



FEATURE ENGINEERING

```
... TFIDF features added and 'productTitle' dropped.

Feature Engineering Completed!
originalPrice price sold tagText discount_percentage adjustable and \
0 30.0 46.79 600 99 -55.966667 0.0 0.0
1 116.0 169.72 9 99 -46.310345 0.0 0.0
2 78.4 39.46 7 99 49.668367 0.0 0.0

bed bedroom bedside ... up upholstered vanity velvet wardrobe \
0 0.0 0.189721 0.0 ... 0.0 0.0 0.0 0.300611
1 0.0 0.000000 0.0 ... 0.0 0.0 0.0 0.000000
2 0.0 0.200469 0.0 ... 0.0 0.0 0.0 0.000000

white wicker with wood wooden
0 0.0 0.000000 0.120182 0.0 0.0
1 0.0 0.300086 0.000000 0.0 0.0
2 0.0 0.000000 0.126991 0.0 0.0

[3 rows x 105 columns]
```

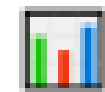
MODEL TRAINING PREDICT NUMBER OF ITEMS SOLD

```
... RandomForestRegressor
RandomForestRegressor(random_state=42)
```

MODEL EVALUTION

 MODEL PERFORMANCE:
Linear Regression → MSE: 1108199.66, R²: -0.208
Random Forest → MSE: 935688.42, R²: -0.020

KEY INSIGHT



INSIGHTS:

1

Total Products: 538

2

Average Price: \$136.75

3

Average Discount: -5428.00%

4

Most Common Shipping Type: 99

EV SALES BY STATES INDIA

SUMMARY

PROJECT OVERVIEW

TITLE: EV SALES BY STATE IN INDIA

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

**DATASET: ALL THE TYPES OF ELECTRICAL VEHICLE SOLD ALL OVER THE
STATES OF INDIA**

PROJECT OBJECTIVE

The goal of the "Electric Vehicle Sales by State in India" project is to analyze and predict electric vehicle (EV) sales trends across different states in India, providing insights into regional adoption patterns, market drivers, and key factors affecting EV uptake. The project aims to help guide strategic decisions for market planning, investment, and infrastructure development in the growing Indian EV sector.

DATASET DETAILS

```
df = pd.read_csv("Electric-Vehicle-Sales-by-State-in-
                India.csv",
                engine='python',      # allows advanced parsing
                on_bad_lines='skip',   # skips lines with parsing
                errors
                encoding_errors='ignore') # ignores text encoding
                errors

                # Confirmation message
                print(" Data Loaded Successfully!\n")

                # Show the shape (rows, columns)
                print("Shape of dataset:", df.shape)

                # Show the first 5 rows
                print("\nPreview:")
                print(df.head())
```

*** Data Loaded Successfully!

Shape of dataset: (96845, 8)

Preview:

	Year	Month_Name	Date	State	Vehicle_Class \
0	2014	jan	01-01-2014	Andhra Pradesh	ADAPTED VEHICLE
1	2014	jan	01-01-2014	Andhra Pradesh	AGRICULTURAL TRACTOR
2	2014	jan	01-01-2014	Andhra Pradesh	AMBULANCE
3	2014	jan	01-01-2014	Andhra Pradesh	ARTICULATED VEHICLE
4	2014	jan	01-01-2014	Andhra Pradesh	BUS

	Vehicle_Category	Vehicle_Type	EV_Sales_Quantity
0	Others	Others	2
1	Others	Others	6
2	Others	Others	9
3	Others	Others	2
4	Bus	Bus	33

METHODOLOGY

1. **Data Collection: Loaded CSV into pandas; inspected shape and types.**
2. **Preprocessing:**
 - **Converted data types**
 - **Encoded categorical variables**
 - **Created new features for time analysis (month/day)**

```
***
Cleaned Data Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 96845 entries, 0 to 96844
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Year                  96845 non-null  Int64
1   Month_Name            96845 non-null  category
2   Date                  96845 non-null  datetime64[ns]
3   State                 96845 non-null  category
4   Vehicle_Class         96845 non-null  category
5   Vehicle_Category      96845 non-null  category
6   Vehicle_Type          96845 non-null  category
7   EV_Sales_Quantity     96845 non-null  int64
dtypes: Int64(1), category(5), datetime64[ns](1), int64(1)
memory usage: 2.8 MB
None

Missing Values Summary:
Year                0
Month_Name          0
Date                0
State               0
Vehicle_Class       0
Vehicle_Category    0
Vehicle_Type        0
EV_Sales_Quantity   0
dtype: int64

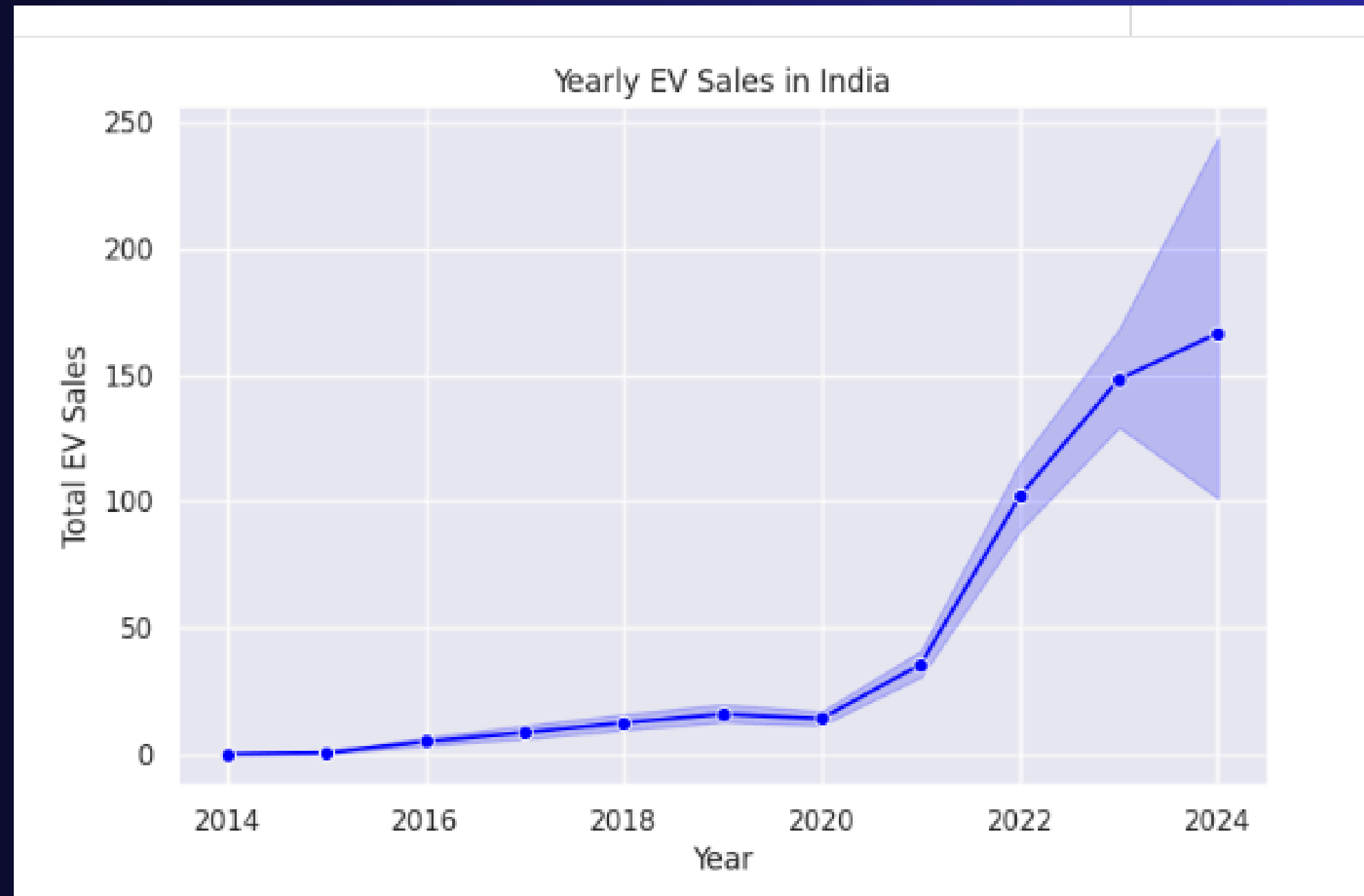
Statistical Summary (Numeric Columns):

```

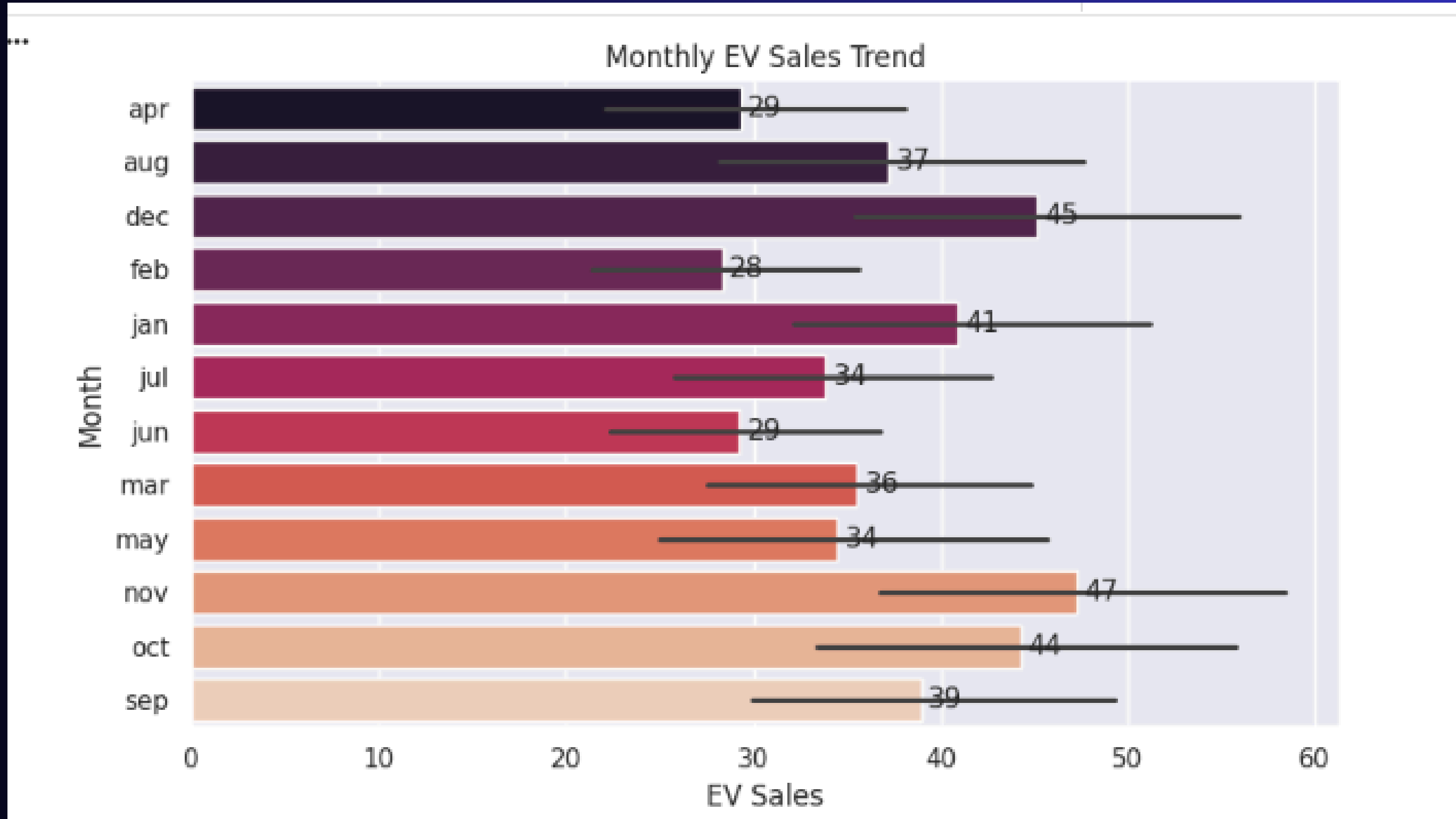
	Year	Date	EV_Sales_Quantity
count	96845.0	96845	96845.000000
mean	2018.622768	2019-01-28 11:00:33.641385728	37.125933
min	2014.0	2014-01-01 00:00:00	0.000000
25%	2016.0	2016-08-01 00:00:00	0.000000
50%	2018.0	2018-07-01 00:00:00	0.000000

DATA VISUALIZTIONS

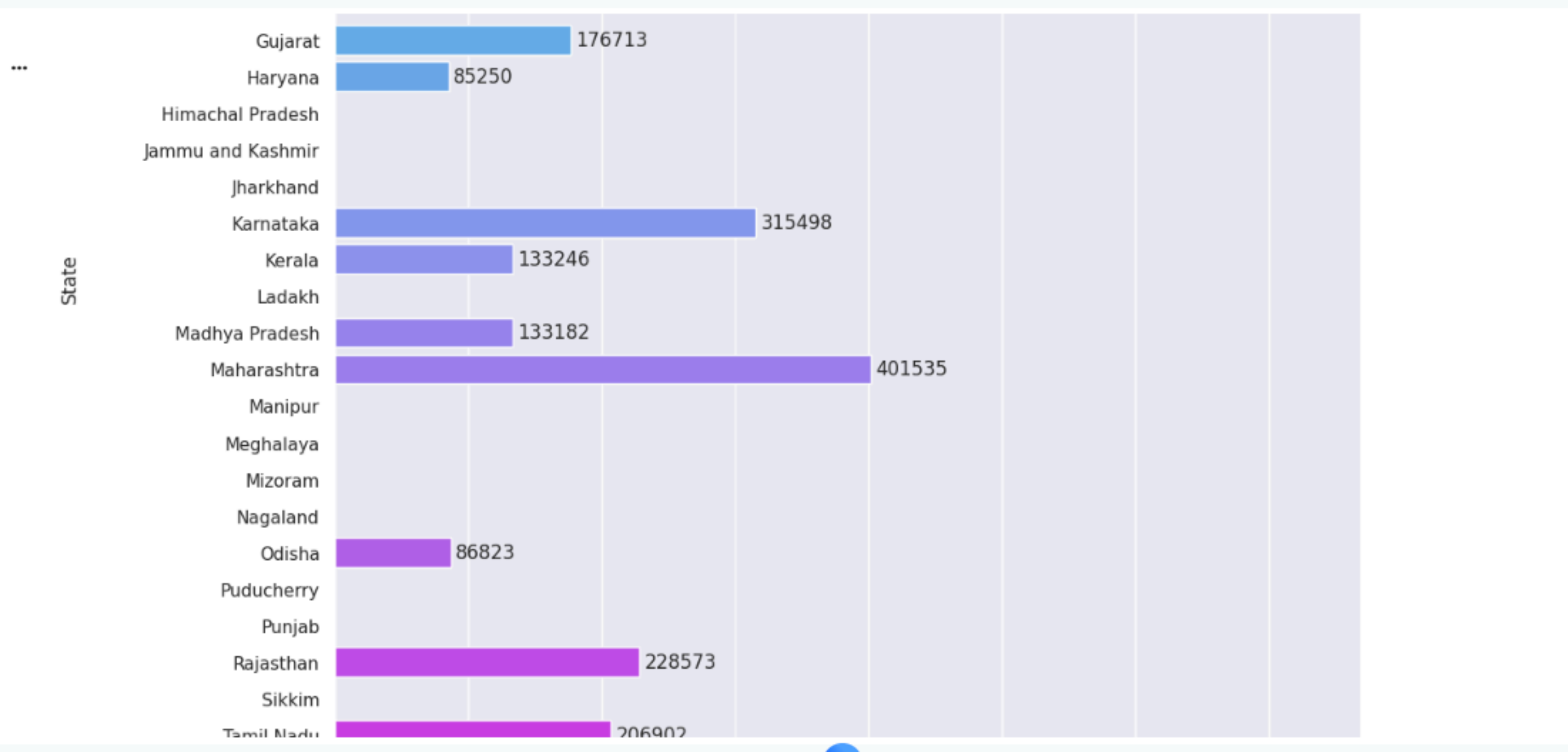
YEARLY TREND OF EV SALES



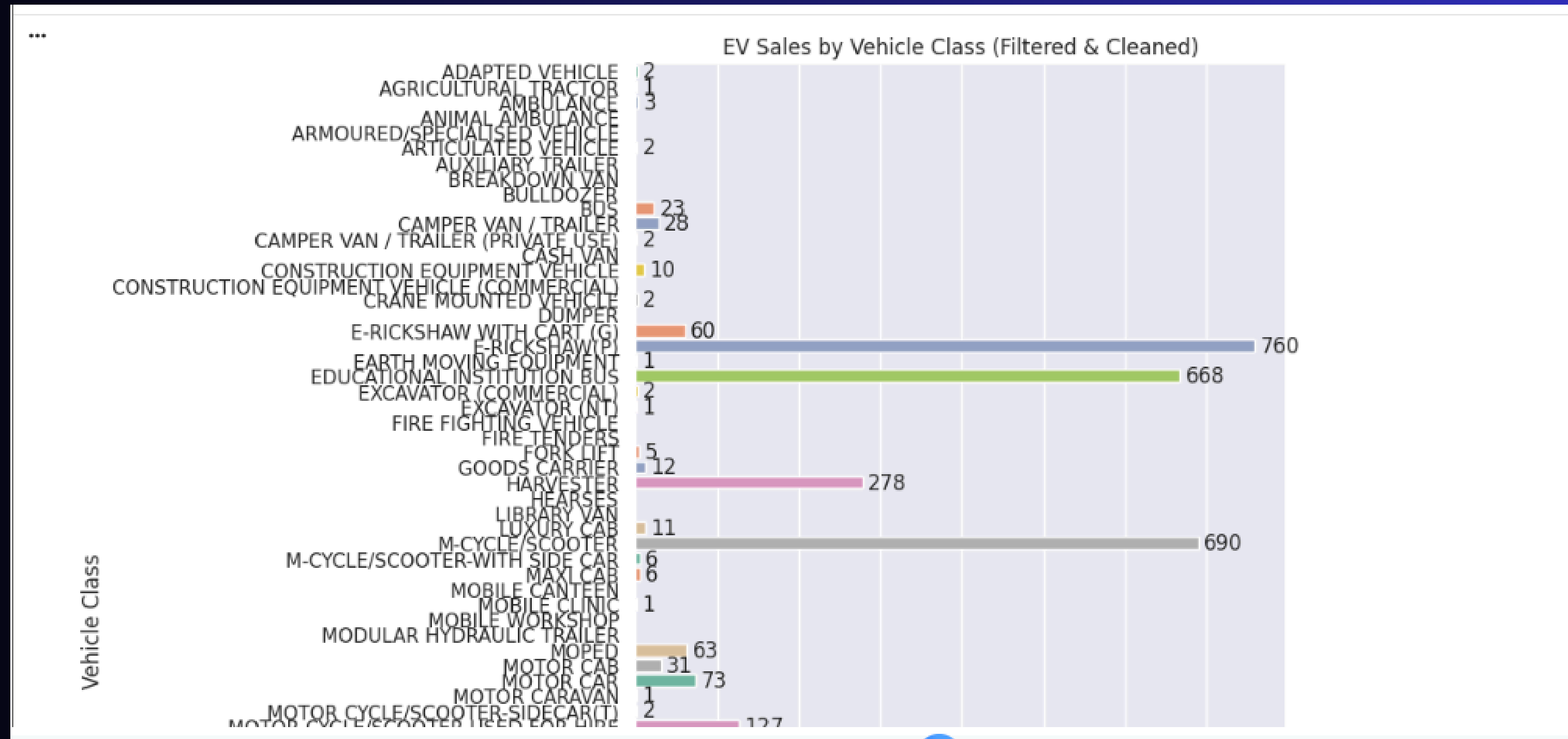
MONTHLY EV SALES TREND



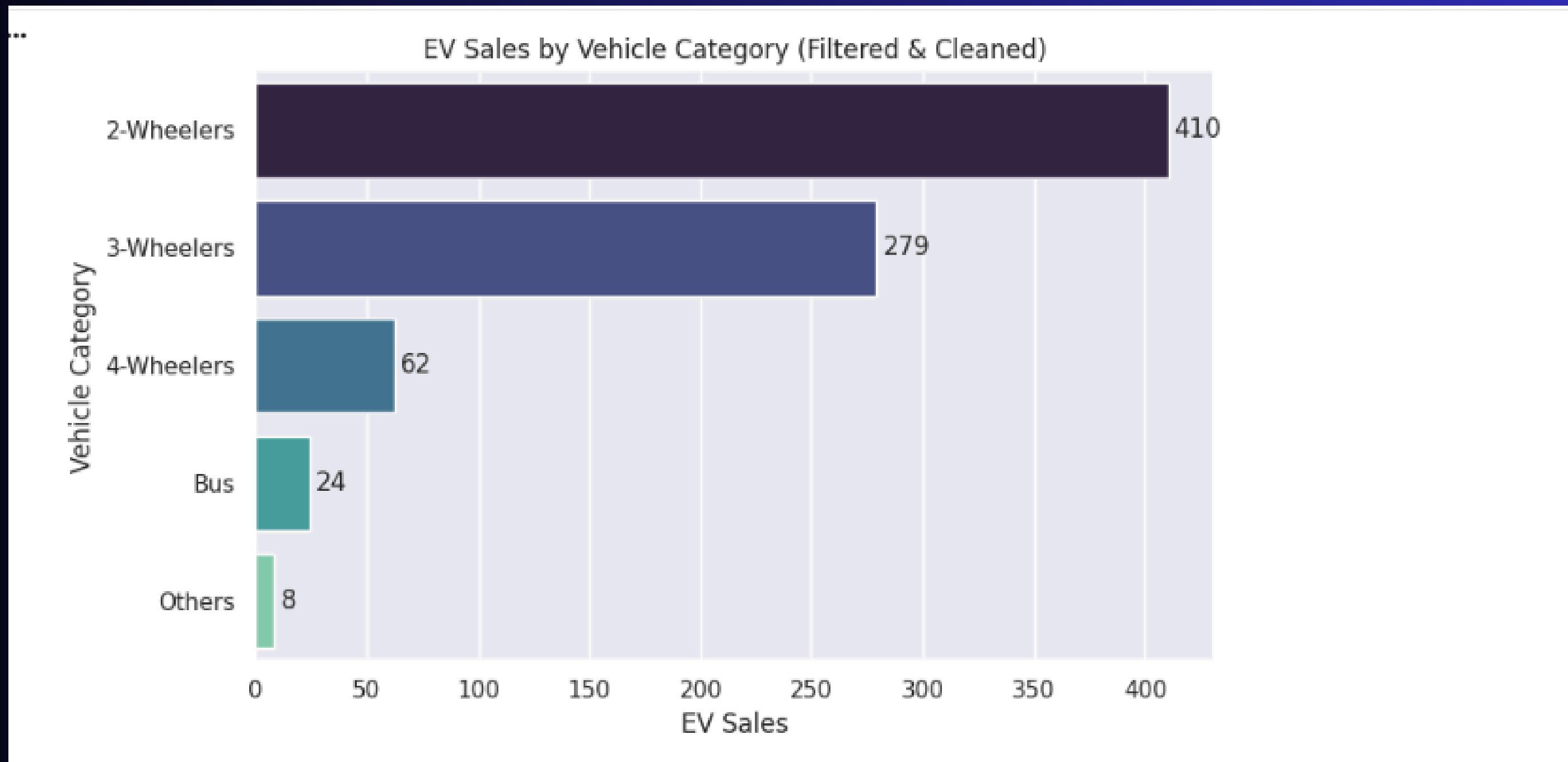
STATE WISE EV SALES



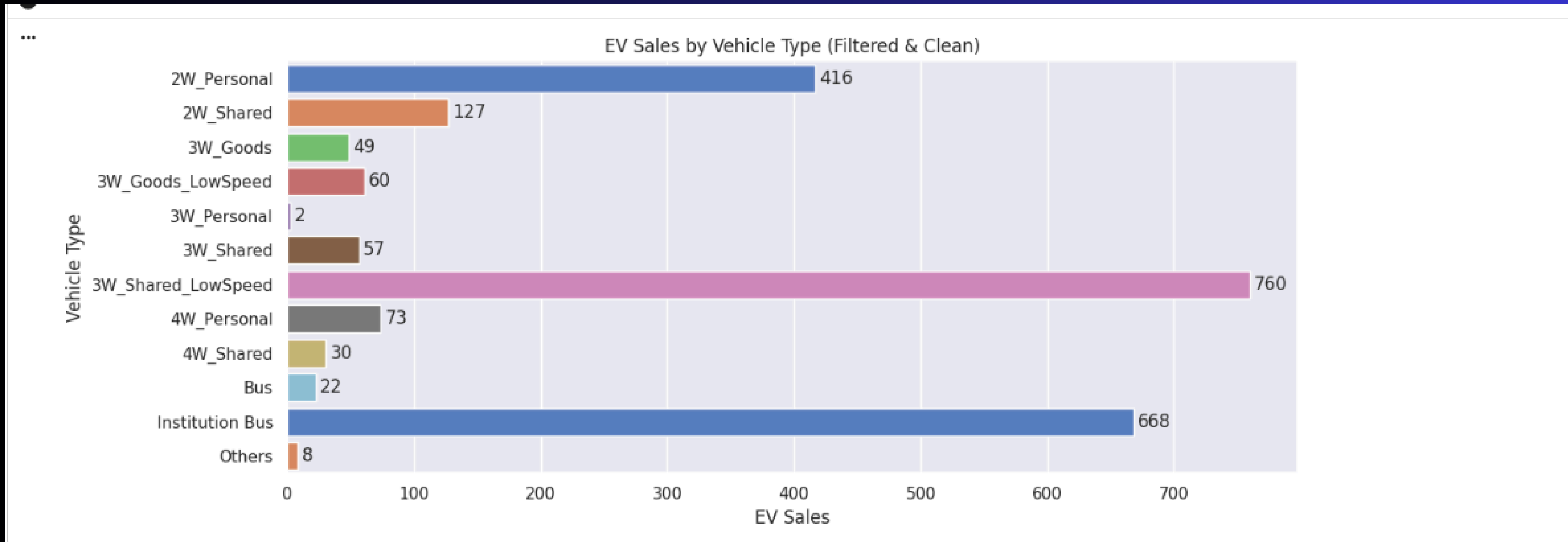
EV SALES BY VEHICLE CLASS



EV SALES BY VEHICLE CATEGORY



EV SALES BY VEHICLE TYPE



KEY INSIGHTS

INSIGHTS:

- 1 Total Records: 96845
- 2 Time Range: 2014 to 2024
- 3 Highest Selling State: Uttar Pradesh (732074 sales)
- 4 Highest Vehicle Category: 2-wheelers (1808105 sales)
- 5 Average EV Sales Quantity: 37.13

GOOGLE PLAY STORE DATA ANALYSIS

SUMMARY

PROJECT OVERVIEW

TITLE: GOOGLE PLAY STORE DATA ANALYSIS

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

**THE DATASET HAS 10,800+ ROWS AND 13 MAIN COLUMNS SUCH AS APP,
CATEGORY, RATING, REVIEWS, SIZE, INSTALLS, TYPE, PRICE, CONTENT RATING,
GENRES, LAST UPDATED, CURRENT VER, AND ANDROID VER.**

PROJECT OBJECTIVE

The objective is to analyze app characteristics (ratings, reviews, size, installs, price, etc.) to find trends, outliers, and patterns that can guide app-business decisions.

DATASET DETAILS

```
apps_df=pd.read_csv("googleplaystore.csv")
reviews_df=pd.read_csv("googleplaystore_user_reviews.csv")
print("App Dataset Shape:",apps_df.shape)
print("User Reviews Dataset Shape:",reviews_df.shape)
apps_df.head()
```

```
*** App Dataset Shape: (10841, 13)
User Reviews Dataset Shape: (64295, 5)
```

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

METHODOLOGY

EXPLORE BASIC INFO

```
Dataset Information
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 10841 entries, 0 to 10840
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   App                    10841 non-null  object
1   Category               10841 non-null  object
2   Rating                 9367 non-null   float64
3   Reviews                10841 non-null  object
4   Size                   10841 non-null  object
5   Installs               10841 non-null  object
6   Type                   10840 non-null  object
7   Price                  10841 non-null  object
8   Content Rating         10840 non-null  object
9   Genres                 10841 non-null  object
10  Last Updated           10841 non-null  object
11  Current Ver            10833 non-null  object
12  Android Ver            10838 non-null  object
dtypes: float64(1), object(12)
memory usage: 1.1+ MB
None

Missing Values
App                0
Category           0
Rating             1474
Reviews            0
Size               0
Installs           0
Type               1
Price              0
Content Rating     1
Genres             0
Last Updated       0
```

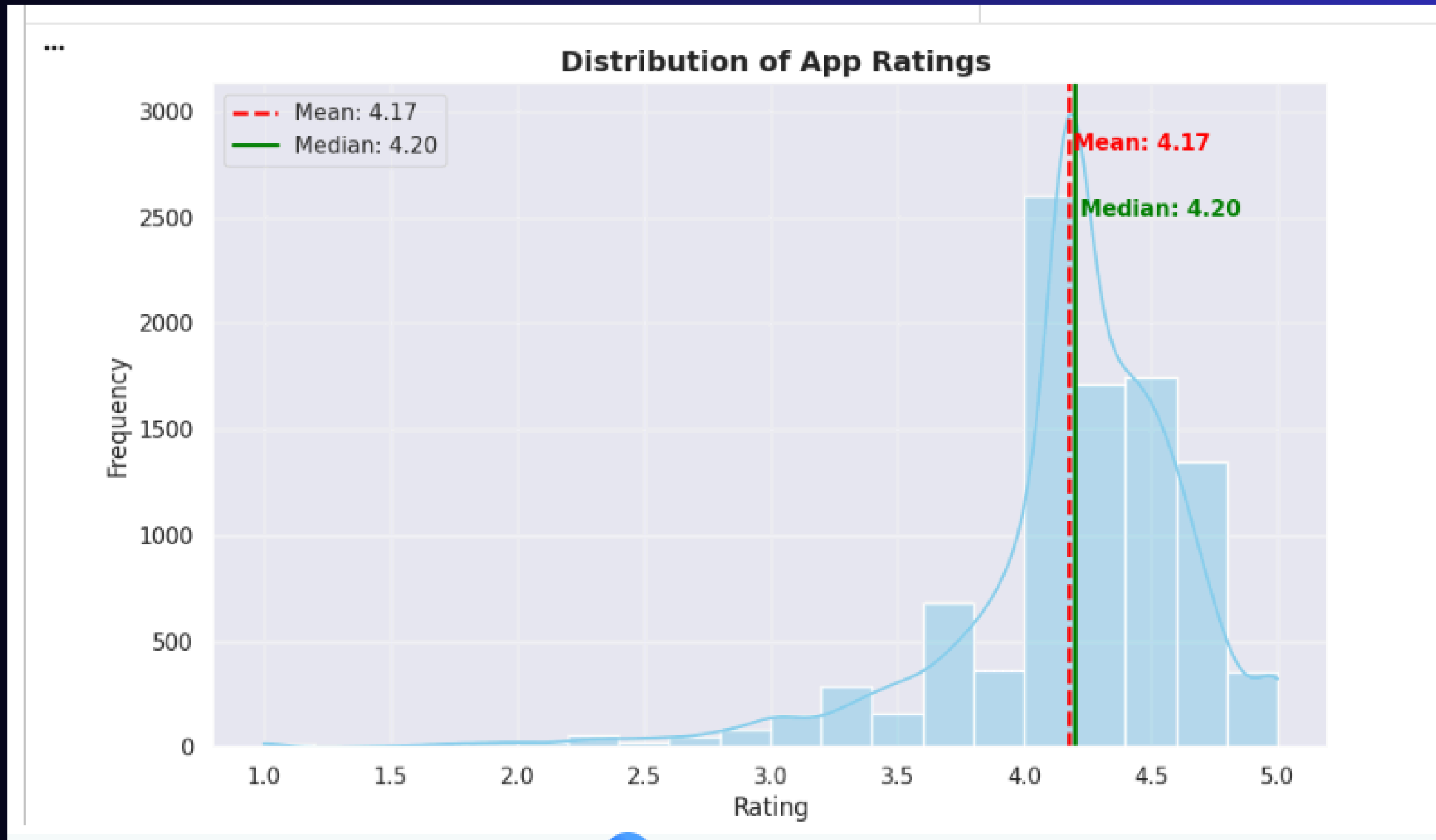
SUMMARY STATISTICS

Summary Of Numeric Columns:

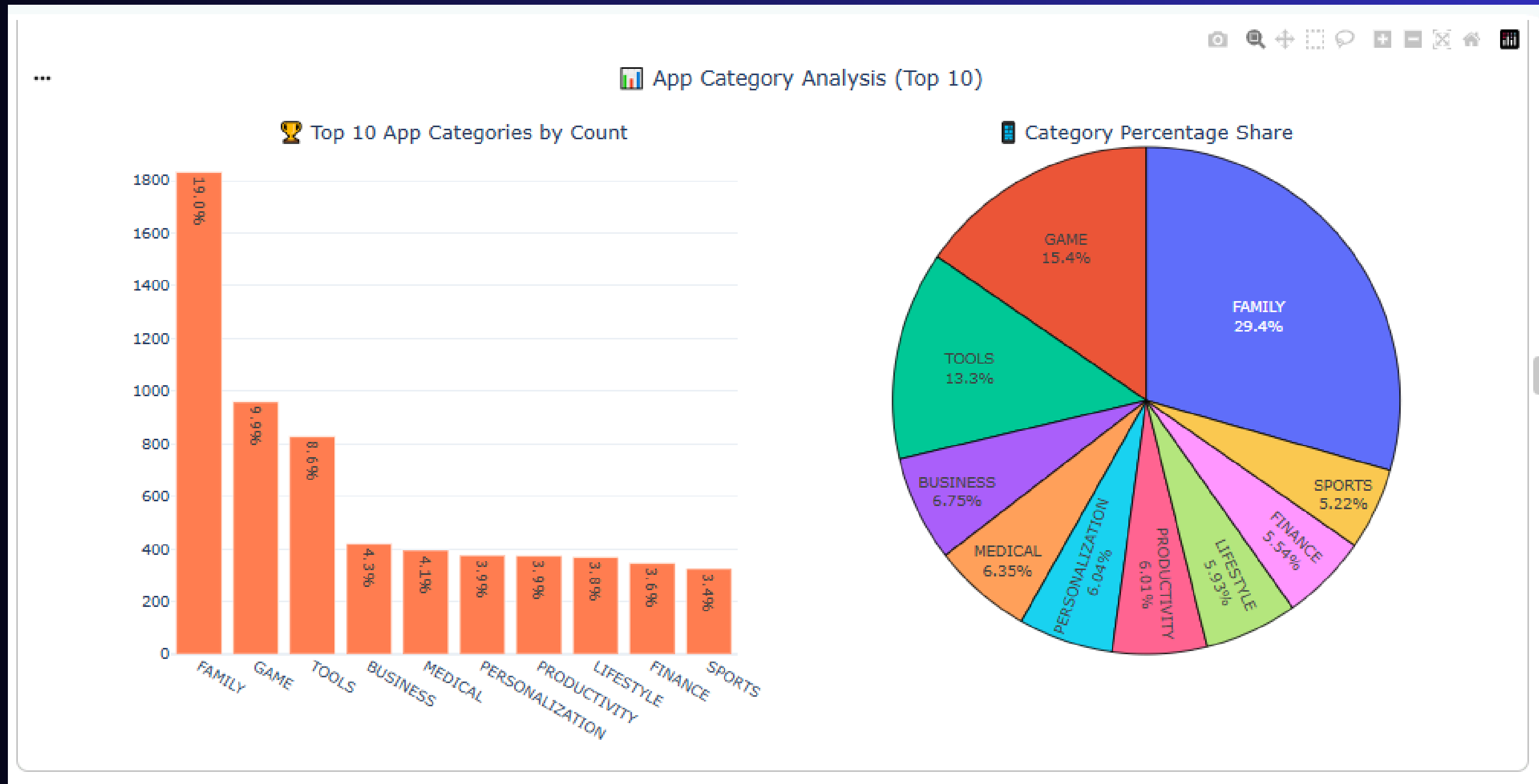
	Rating	Reviews	Installs	Price	Size_KB
count	9659.000000	9.659000e+03	9.659000e+03	9659.000000	8432.000000
mean	4.173517	2.165926e+05	7.777507e+06	1.099299	20884.374134
std	0.494313	1.831320e+06	5.375828e+07	16.852152	22351.767303
min	1.000000	0.000000e+00	0.000000e+00	0.000000	8.500000
25%	4.000000	2.500000e+01	1.000000e+03	0.000000	4710.400000
50%	4.200000	9.670000e+02	1.000000e+05	0.000000	12288.000000
75%	4.500000	2.940100e+04	1.000000e+06	0.000000	28672.000000
max	5.000000	7.815831e+07	1.000000e+09	400.000000	102400.000000

DATA VISUALIZTIONS

DISTRIBUTION OF APPS RATING



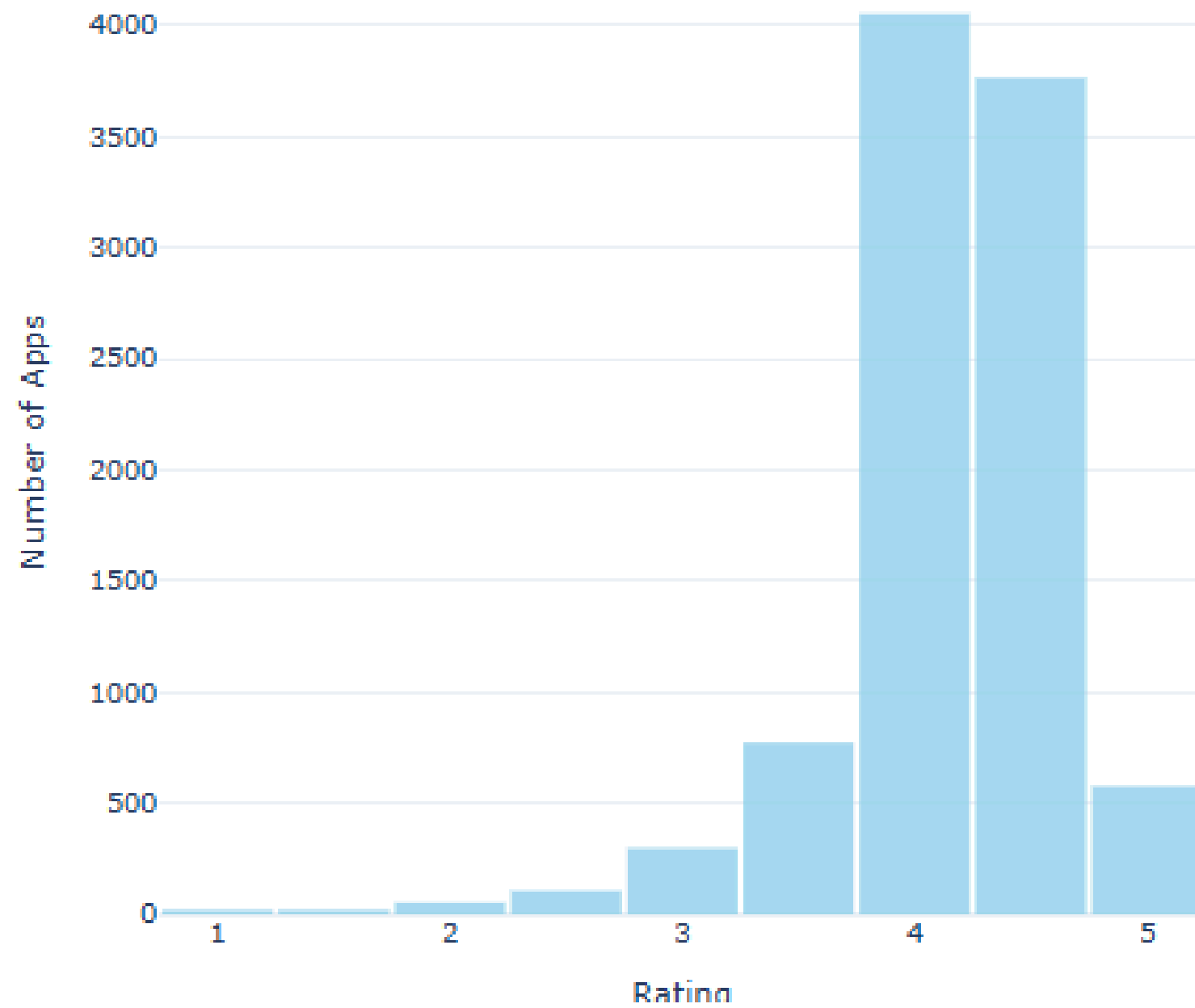
APP CATEGORIES (TOP 10)



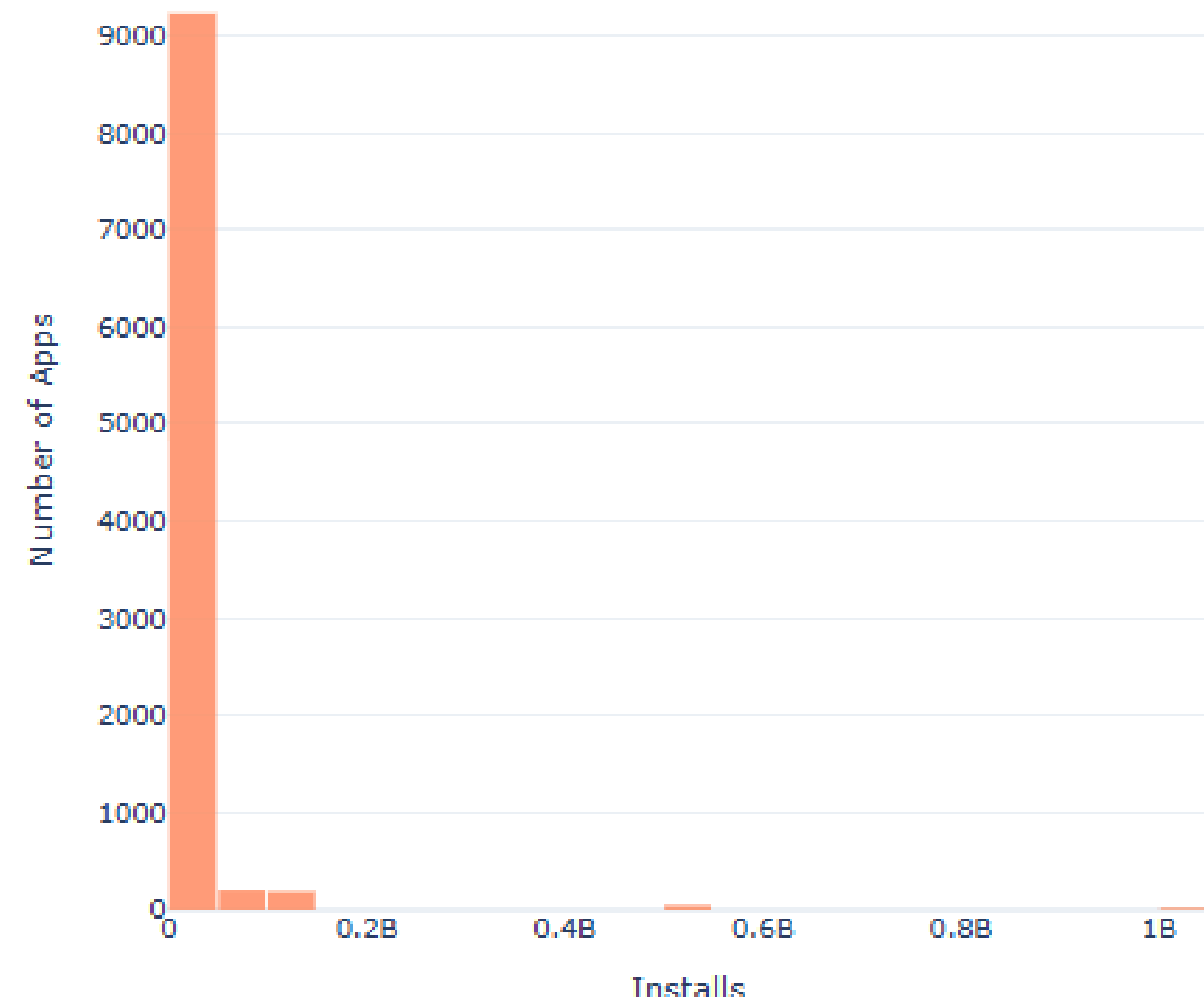
RATING VS INSTALL

Comparison of App Ratings and Installs

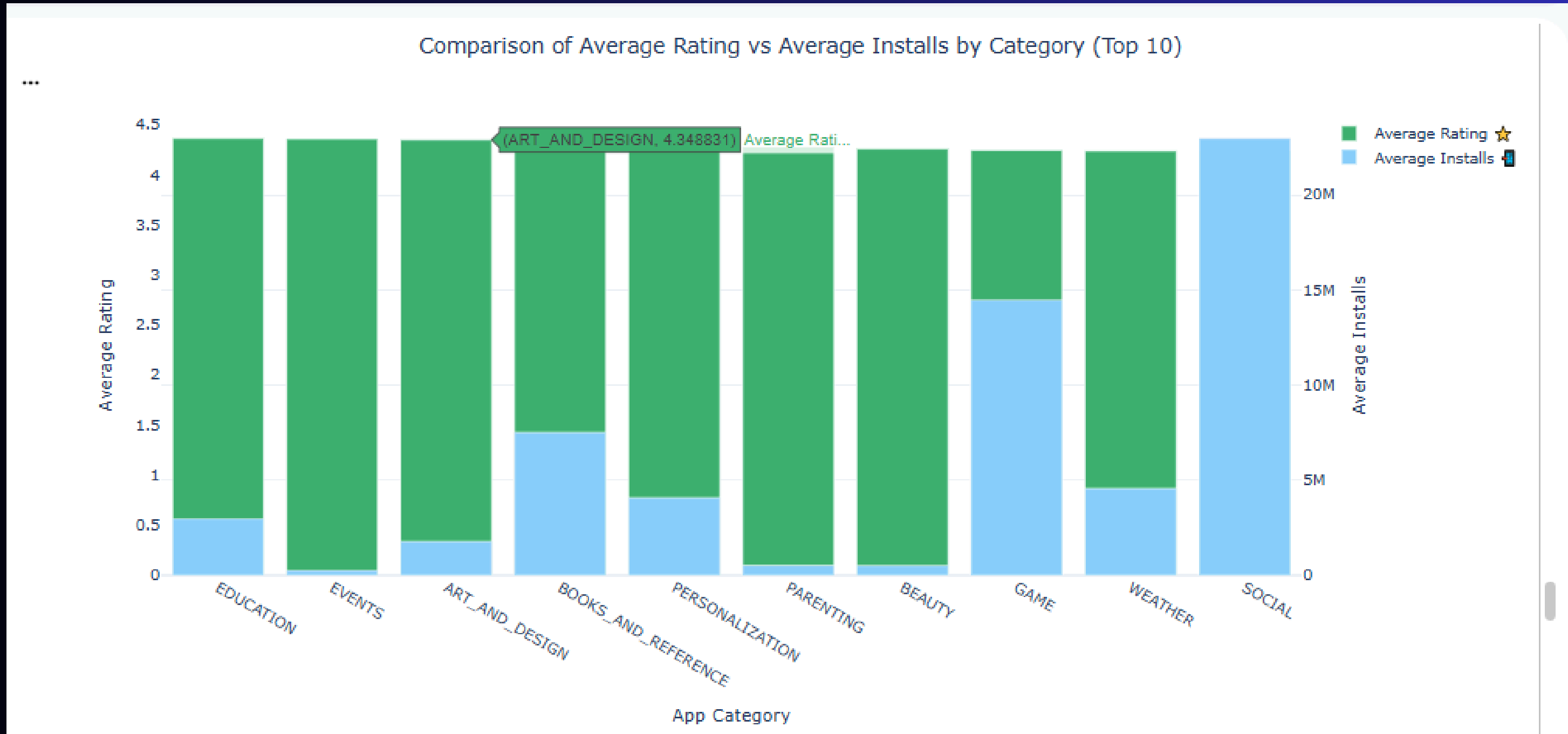
Distribution of App Ratings



Distribution of App Installs

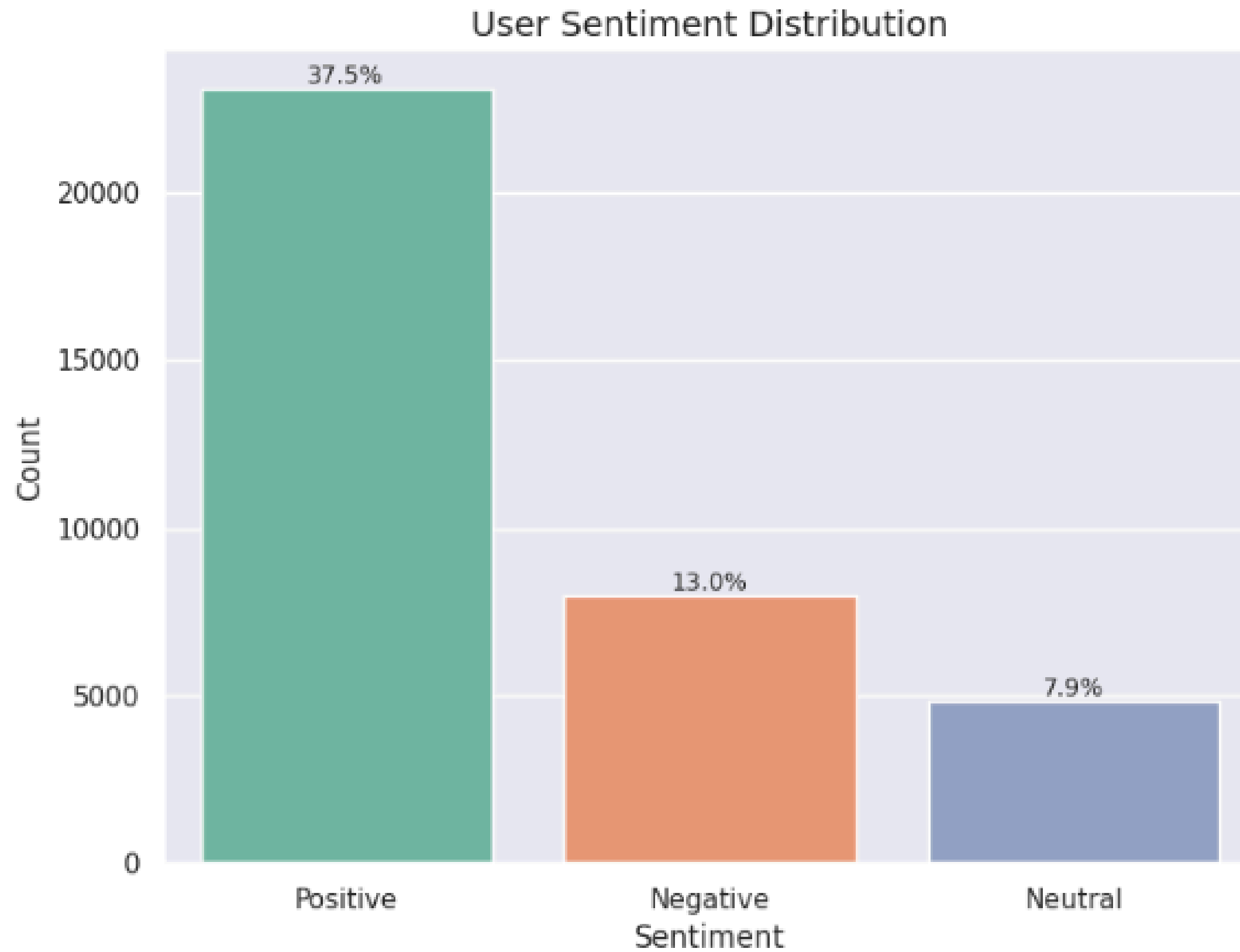


AVERAGE RATING



THE USER REVIEW

4 coloring book found Positive



KEY INSIGHTS

...		App Dataset — Key Insights	
		Metric	Value
		Total Apps Analyzed	9,659
		Average App Rating	4.170000
		Median App Rating	4.200000
		Most Popular Category	FAMILY
		Most Installed App	Google Play Books
		Most Installed App (Installs)	1,000,000,000
		Total Installs (All Apps)	75,122,937,527

SUMMER OLYMPIC 1976- 2008

SUMMARY

PROJECT OVERVIEW

TITLE: SUMMER OLYMPIC 1976-2008

DOMAIN: DATA ANALYST

**TOOL USED: PYTHON, PANDAS, SCIKIT-LEARN, MATPLOTLIB, SEABORN,
EXCEL**

PROJECT OBJECTIVE

Analyze Olympic medal trends from 1976–2008.

Identify top-performing countries, athletes, and sports.

Explore gender participation patterns.

**Build a machine learning model to predict medal outcomes
based on attribute**

DATASET DETAIL

```
data=pd.read_csv("Summer-Olympic-medals-1976-  
to-2008.csv",encoding="latin1")  
print("DATASET LOADED SUCCESSFULLY ")  
print("SHAPE OF DATASET",data.shape)  
print("\nPreview:")  
print(data.head())
```

```
*** DATASET LOADED SUCCESSFULLY  
SHAPE OF DATASET (15433, 12)  
  
Preview:  
      City  Year  Sport Discipline  Event \  
0  Montreal  1976.0  Aquatics    Diving  3m springboard  
1  Montreal  1976.0  Aquatics    Diving  3m springboard  
2  Montreal  1976.0  Aquatics    Diving  3m springboard  
3  Montreal  1976.0  Aquatics    Diving  3m springboard  
4  Montreal  1976.0  Aquatics    Diving  10m platform  
  
      Athlete Gender Country_Code  Country  Event_gender \  
0      KÖHLER, Christa  Women      GDR  East Germany      W  
1      KOSENKOV, Aleksandr  Men      URS  Soviet Union      M  
2      BOGGS, Philip George  Men      USA  United States      M  
3  CAGNOTTO, Giorgio Franco  Men      ITA      Italy      M  
4      WILSON, Deborah Kepler  Women      USA  United States      W  
  
      Medal  Count  
0  Silver   NaN  
1  Bronze   NaN  
2   Gold   NaN  
3  Silver   NaN  
4  Bronze   NaN
```

Methodology

DATA CLEANING

... Checking for missing values:

City	117
Year	117
Sport	117
Discipline	117
Event	117
Athlete	117
Gender	117
Country_Code	117
Country	117
Event_gender	117
Medal	117
Count	15433
dtype: int64	



Cleaned Data:

<class 'pandas.core.frame.DataFrame'>

Index: 15316 entries, 0 to 15432

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	City	15316 non-null	object
1	Year	15316 non-null	int64
2	Sport	15316 non-null	object
3	Discipline	15316 non-null	object
4	Event	15316 non-null	object
5	Athlete	15316 non-null	object
6	Gender	15316 non-null	object
7	Country	15316 non-null	object
8	Medal	15316 non-null	object
9	Count	0 non-null	float64

dtypes: float64(1), int64(1), object(8)

memory usage: 1.3+ MB

DATA VISUALIZATION

CITIES THAT HOSTED OLYMPICS

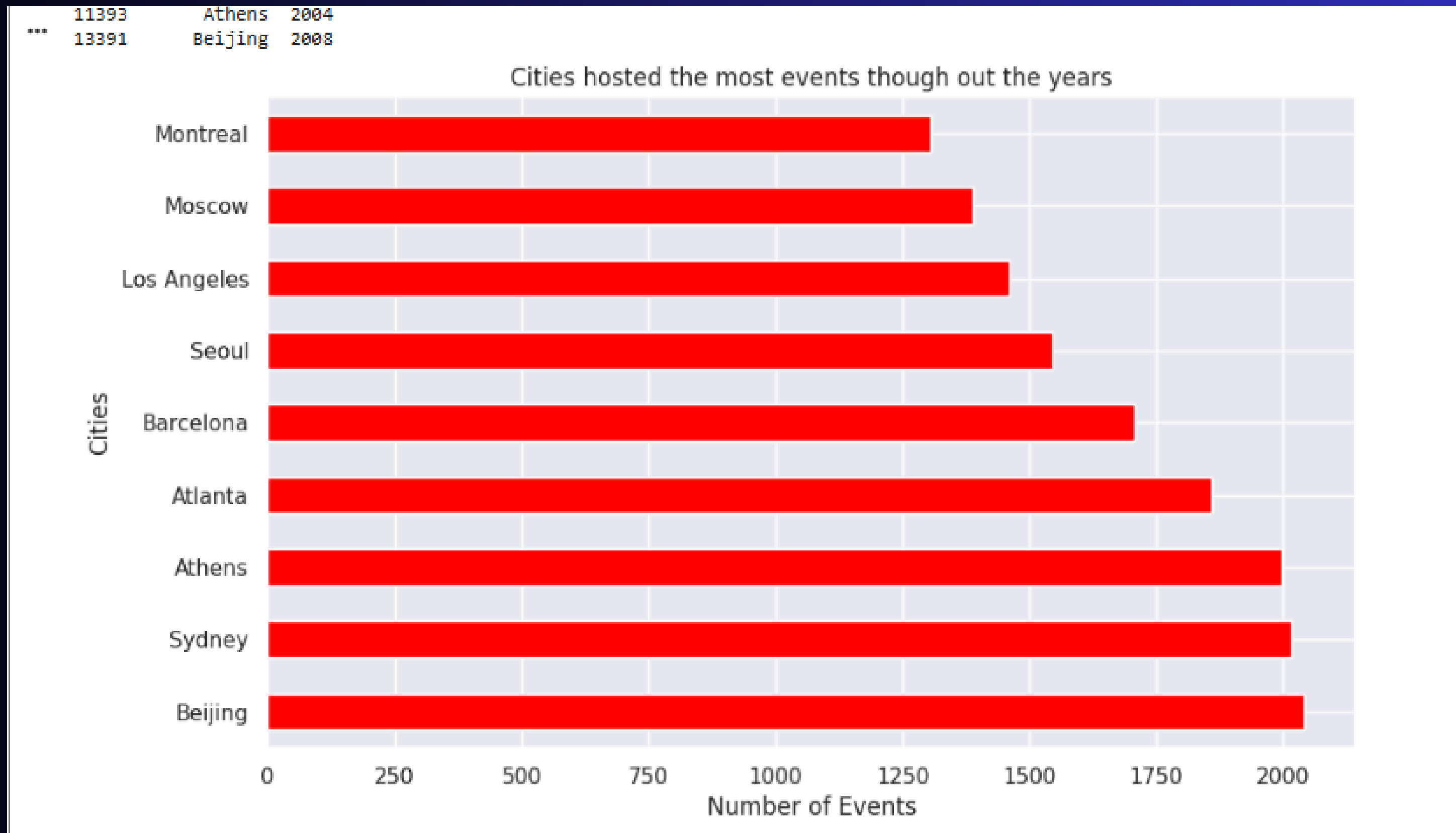
=====

CITIES THAT HOSTED THE OLYMPICS (1976-2008)

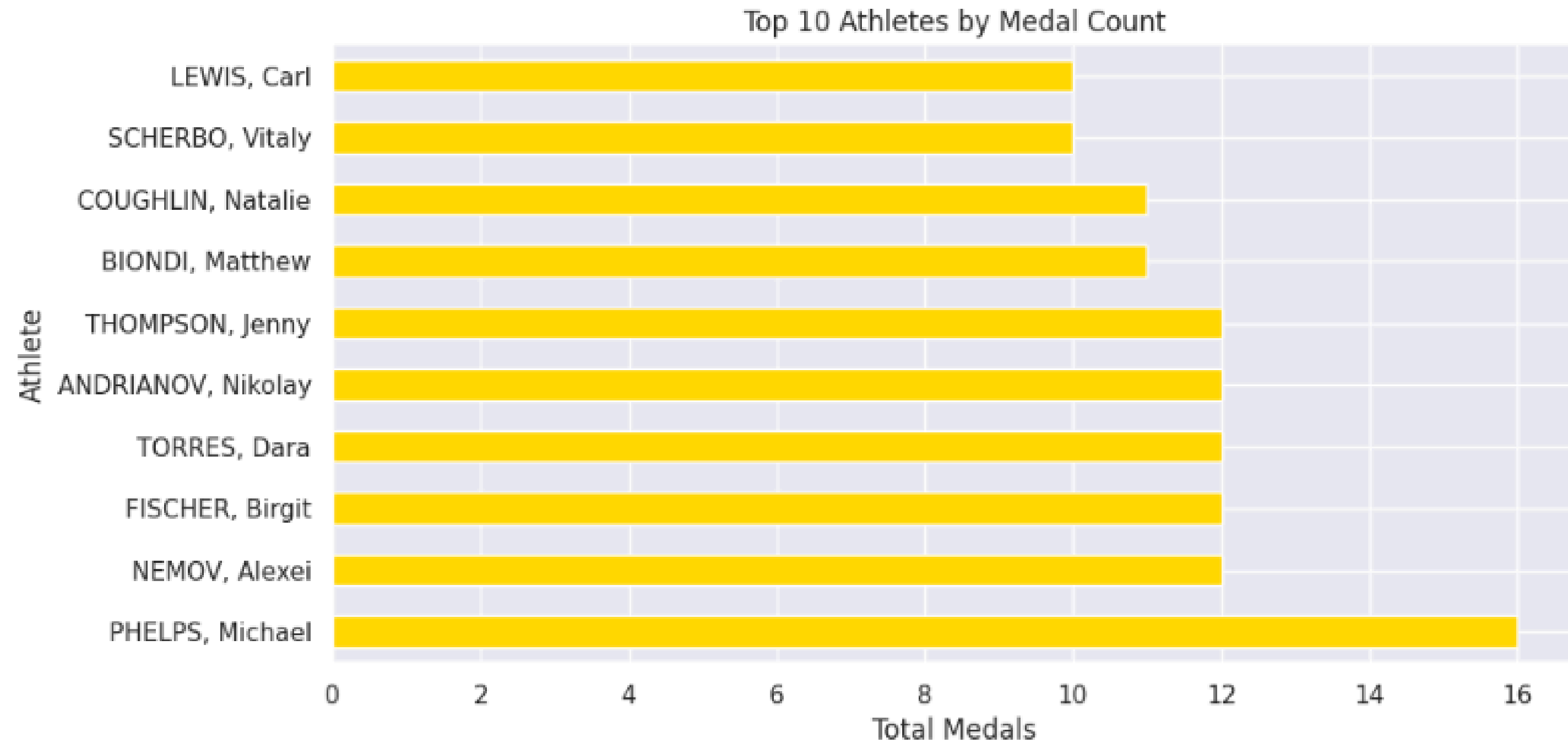
=====

	City	Year
0	Montreal	1976
1422	Moscow	1980
2809	Los Angeles	1984
4268	Seoul	1988
5814	Barcelona	1992
7519	Atlanta	1996
9378	Sydney	2000
11393	Athens	2004
13391	Beijing	2008

WHICH CITIES HOSTED THE MOST EVENTS

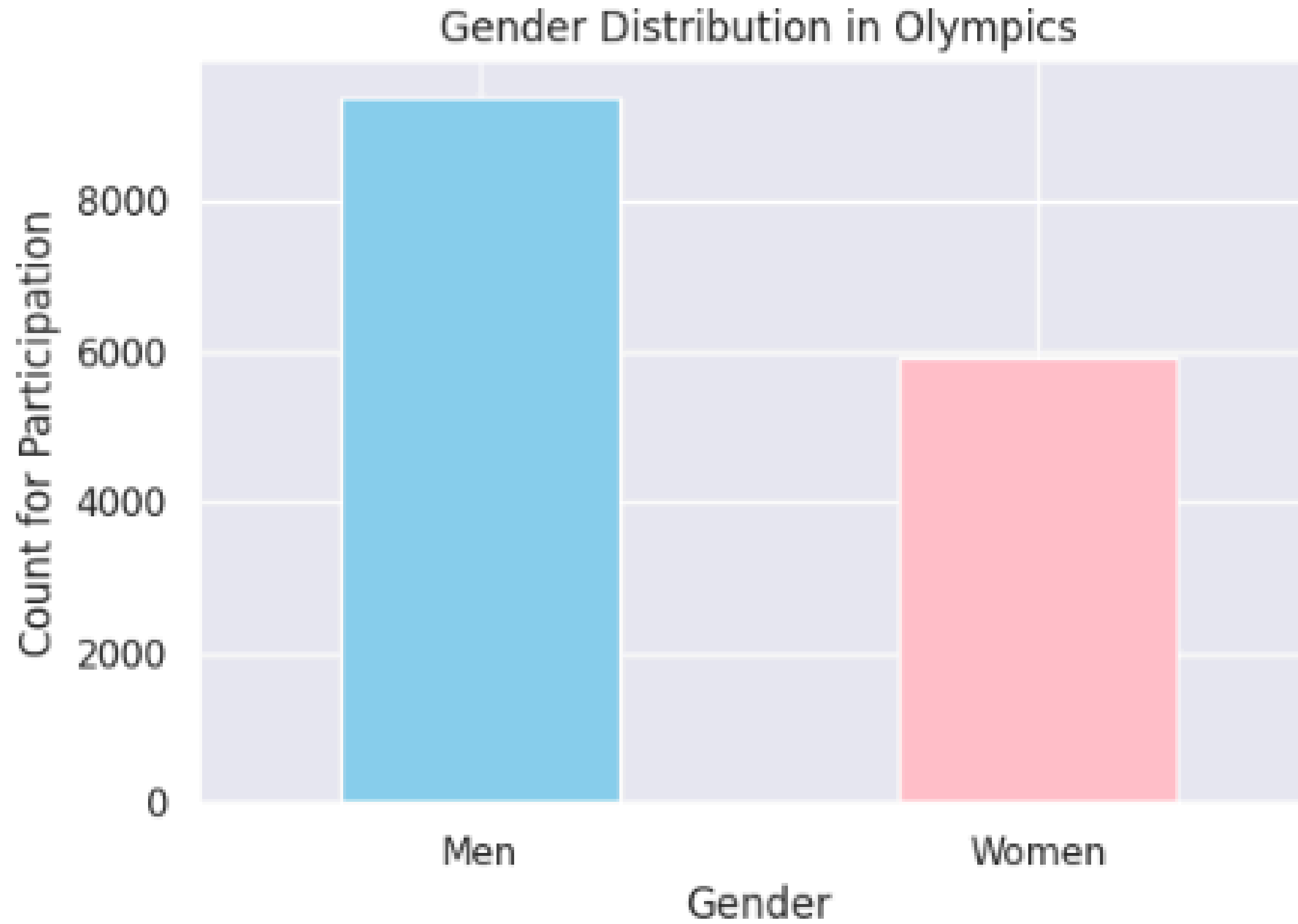


WHICH ATHLETE WON THE MOST MEDALS



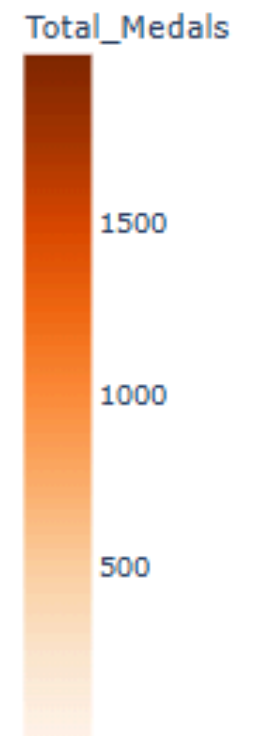
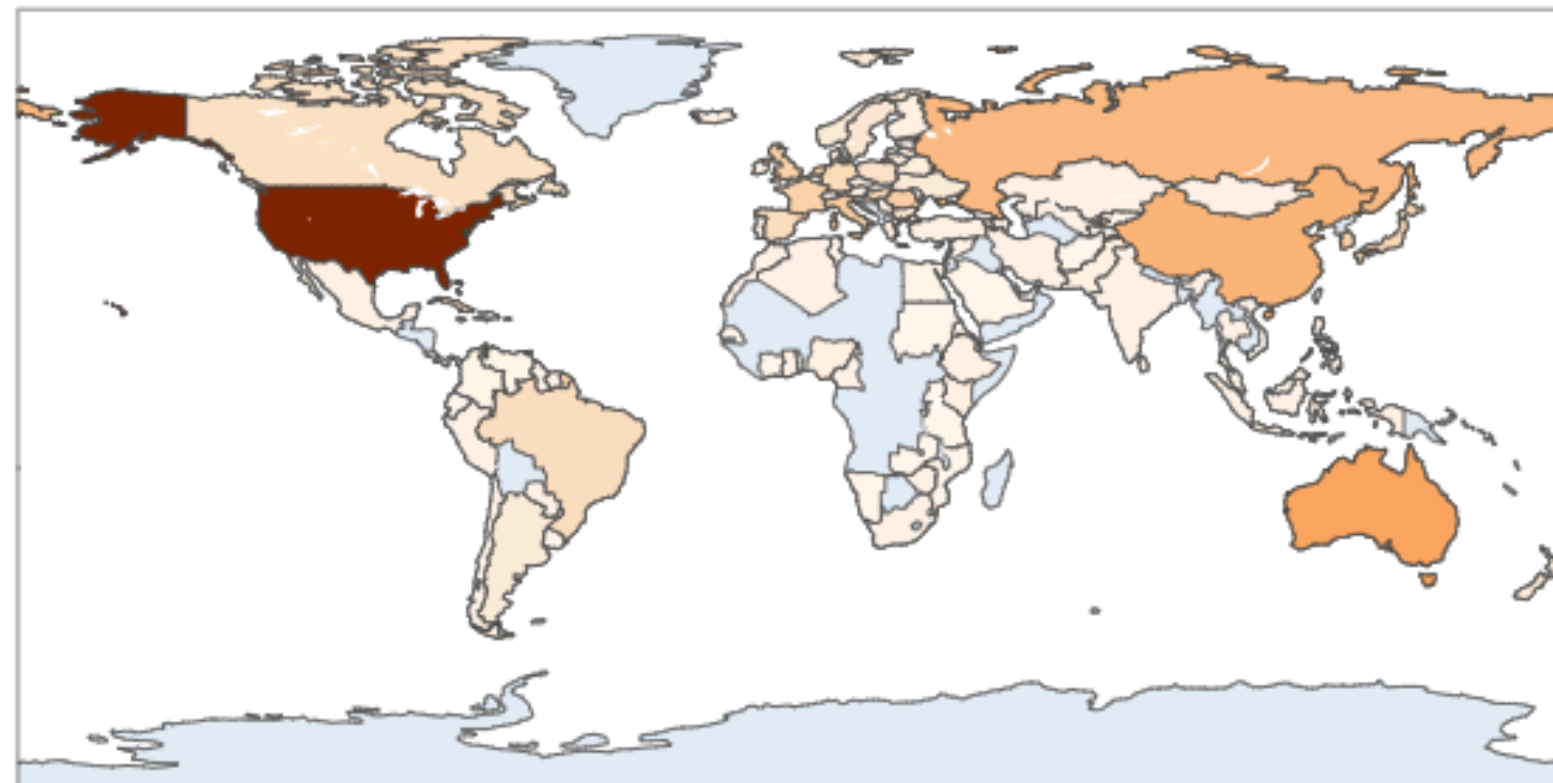
GENDER DISTRIBUTION

...



TOTAL MEDAL PER COUNTRY

Total Olympic Medals by Country (1976–2008)



GOLD MEDALS MAP

Gold Medal Distribution (1976–2008)

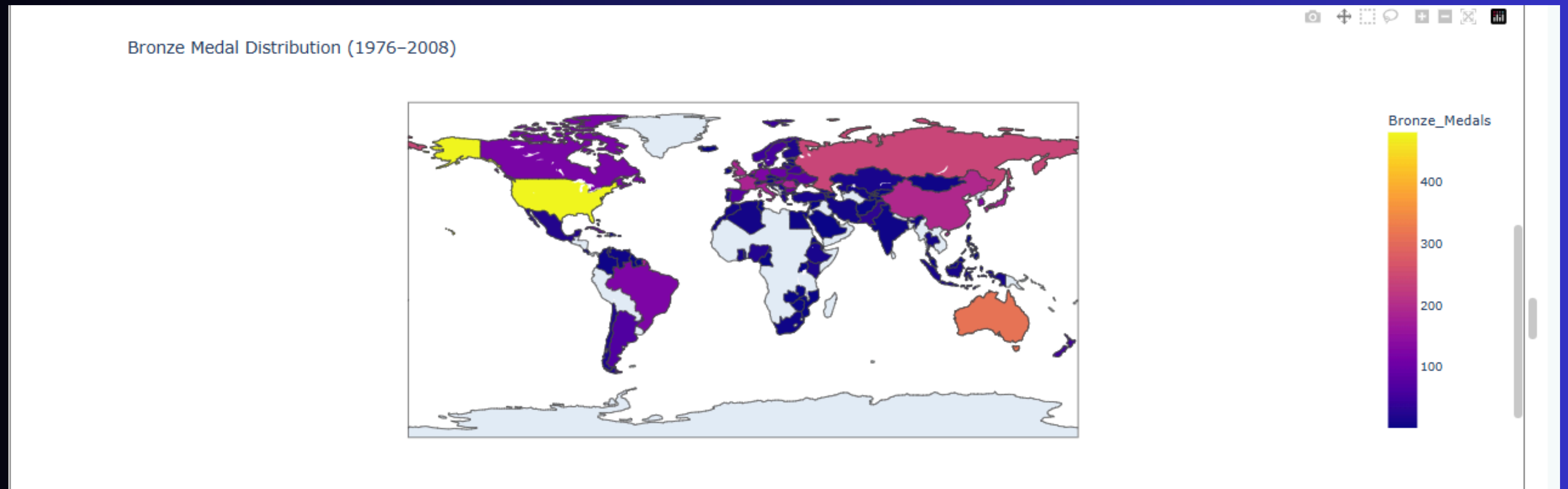


SILVER MEDALS MAP

Silver Medal Distribution (1976–2008)

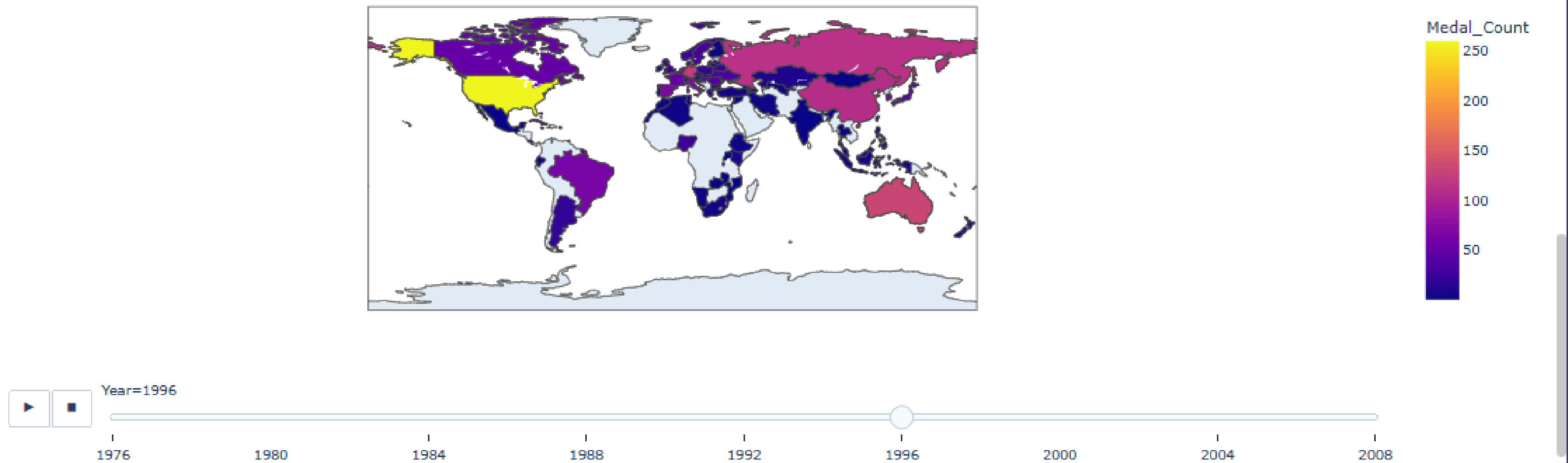


BRONZE MEDALS MAP



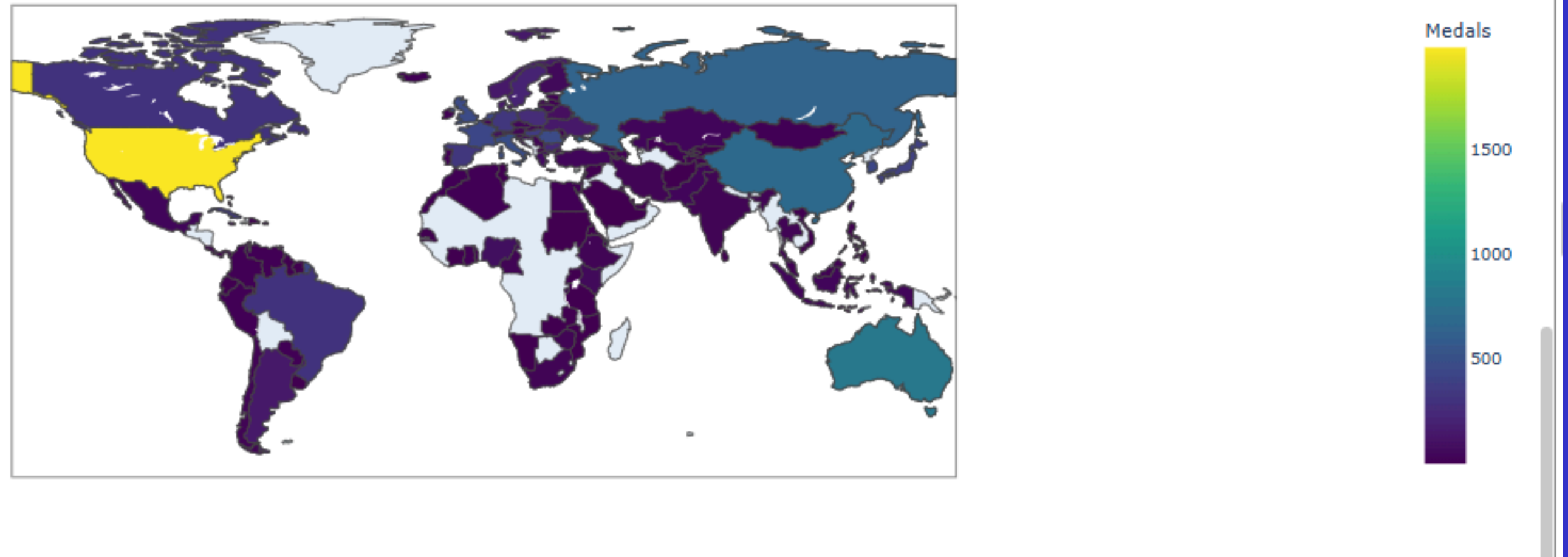
MEDALS OVER YEARS

Olympic Medals Over the Years (1976-2008)



TREND OF MEDALS OVER THE YEARS

World Map: Medal Distribution



KEY INSIGHT

=====

KEY INSIGHTS FROM OLYMPICS DATA (1976-2008)

=====

1 COUNTRY PERFORMANCE

- United States secured the highest medal count (1992 medals).
- A small number of countries dominate the majority of medals.

2 GENDER PARTICIPATION

- Men athletes have participated more in the tournament.
- Female participation, although lower, shows steady improvement.

3 ATHLETE PERFORMANCE

- PHELPS, Michael is the most decorated athlete with 16 medals.
- Very few athletes win medals consistently across multiple years.

4 HOSTING CITY OBSERVATIONS

- Beijing hosted the most Olympic events (2042 times).
- Hosting duties rotate widely among global cities.

5 YEARLY MEDAL TREND

- Highest number of medals were awarded in 2008 (2042 medals).
- Medal counts show fluctuations based on participation and event count.

6 GEOGRAPHICAL DISTRIBUTION

- North America, Europe, and parts of Asia lead the medal tally.
- Developing nations show lower medal counts due to resource and training gaps.

=====

✓ END OF KEY INSIGHTS

=====

THANK

YOU